AI-INTEGRATED MULTI-MODAL ACADEMIC EVALUATION, PERFORMANCE INTELLIGENCE, AND AUTOMATED VIVA ASSESSMENT SYSTEM

COMMON INTEGRATED SOLUTION REPORT

Gradex LMS | R26-SE-025

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Gradex is an integrated academic assessment platform that connects handwritten answer grading, handwritten entity relationship diagram evaluation, availability-aware viva scoring, and assessment-driven practice recommendation. The project addresses a common problem: heterogeneous assessment evidence is difficult to convert into consistent, reviewable feedback and useful follow-up learning activities. Its four components combine optical character recognition, rubric-scoped language models, object detection and structural reconstruction, deterministic multimodal scoring, and cognitive and topic analytics within a shared lecturer workflow. Evaluation combines synthetic stage tests, a small diagram validation set, reported scoring simulations, and a repository-based recommendation case study. The handwritten component returned in-course context for 17 of 18 relevant queries and rejected all 54 out-of-scope queries. Diagram detection reported 88.4% precision and 61.3% mean average precision at an intersection-over-union threshold of 0.5 on five validation images. Viva scoring was evaluated using 20,000 sampled feature vectors across thirteen missingness conditions. The reported results show that availability-aware renormalisation reduced mean absolute score disturbance from 9.12 to 3.04 points and mean grade-band changes from 65.8% to 29.5%, compared with zero-imputation. Under a separate correlated-degradation scenario, disturbance was 2.82 versus 9.09 points, and grade-band changes were 26.3% versus 62.6%. The analytics classifier matched all eleven fixture labels, but the five-signal recommendation rule did not demonstrate additional value over weakness-only ranking. These findings support selected engineering mechanisms rather than the validity of autonomous academic grading. Human-marker agreement, independent stakeholder evaluation, and a controlled test spanning all four components remain outstanding. The integrated contribution is a reviewable assessment-to-practice architecture that retains component provenance, distinguishes unavailable evidence from poor performance, and separates numerical assessment from generated advice. Future work should prioritise authentic-data validation, shared status semantics, independent evaluation, and reliable deployment before marks are used consequentially.

Keywords: automated assessment, multimodal evidence, rubric grading, score integrity, learning analytics

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We appreciate the encouragement of our families and peers during the research process. Each member remains responsible for the accuracy of the component evidence attributed to them, and the group accepts responsibility for reviewing the common interpretation, references, and conclusions in this report.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
BT	Bloom taxonomy level
CNN	Convolutional neural network
DBMS	Database management systems
ER	Entity relationship
HNR	Harmonics-to-noise ratio
IoU	Intersection over union
JSON	JavaScript Object Notation
LLM	Large language model
LoRA	Low-rank adaptation
mAP	Mean average precision
NDCG	Normalised discounted cumulative gain
OCR	Optical character recognition
P@5	Precision at five
RAG	Retrieval-augmented generation
SDG	Sustainable Development Goal
WPM	Words per minute

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Assessment in computing education involves more than calculating a final mark. Students explain concepts in handwritten answers, represent database structures in diagrams, and defend reasoning during oral examinations. Lecturers must interpret these different forms against assessment criteria, associate every result with the correct student and course, and provide feedback that can guide subsequent learning. The workload includes recognition, judgement, administration, moderation, and interpretation. Automating one of these activities without preserving the surrounding context can move effort elsewhere or make an error harder to identify.

The Gradex project, R26-SE-025, investigates how these activities can be supported through one modular platform. Its four research components concern handwritten short-answer grading [1], handwritten entity relationship diagram evaluation [2], robust scoring of viva delivery under missing measurements [3], and examination analytics with weakness-aligned practice recommendation [4]. Each component addresses a different representation of evidence. Their common purpose is to make the path from student work to feedback and follow-up practice explicit and reviewable.

Automatic short-answer grading has developed from manually specified concepts and lexical comparison towards learned representations and language models. The literature surveyed in the handwritten component identifies a continuing difference between scoring already digitised text and processing a photographed script from beginning to end [1], [5]. Recognition errors and misplaced answer boundaries can undermine an otherwise plausible grading response. A convincing justification therefore cannot be judged independently of the transcript and question to which it was attached.

Diagram assessment adds a structural requirement. The words in an entity relationship diagram do not express its complete meaning: connections determine which attributes belong to an entity and how entities participate in relationships. The diagram component consequently reconstructs a representation from detected objects, labels,

and lines before rubric evaluation [2]. Its significance within the group system is that graphical evidence cannot safely be treated as a failed instance of text recognition.

Viva assessment introduces a further distinction between academic knowledge and delivery. A recording can support analysis of speech and visible behaviour, but unavailable measurements should not automatically lower a candidate's score. Equally, a measurable signal is not necessarily a valid indicator of technical competence. The viva component addresses the narrower engineering question of how a permitted behavioural score should respond to partial observability [3]. Its policy remains separate from an external technical-accuracy score.

Assessment analytics closes the feedback cycle. Two students with the same overall percentage may need different practice because their topic coverage and cognitive demands differ. Learning analytics provides a basis for organising such evidence [6]. Gradex combines deterministic mark aggregation with topic and Bloom-level interpretation, then ranks candidate practice items [4]. This produces an assessment-to-practice workflow while leaving the educational value of the recommendations open to evaluation.

1.2 Stakeholders and user needs

Lecturers and examiners are the primary operational users. They need to configure courses and rubrics, verify student identities, inspect extracted evidence, review proposed marks, and explain decisions. Their central requirement is not simply a faster numerical output. It is a result whose source, assumptions, and failure conditions can be inspected before approval. The handwritten component's roster validation, per-question engine labels, and override workflow address parts of this requirement [1].

Students are the principal affected stakeholders. A wrong question boundary can attribute their words to the wrong criterion; an unreadable recording can be mistaken for poor delivery; and a misleading recommendation can direct study towards an unsuitable topic. Students therefore need accurate attribution, comprehensible feedback, an accessible route to review, and clarity about the difference between an automated proposal and an approved academic result. The current evidence does not establish that these needs have been validated through a representative student study.

Course coordinators and academic administrators need comparable evidence across a cohort, but comparability requires stable rubrics and explicit denominators. A percentage computed from available marks can conceal missing questions unless coverage accompanies the value. Likewise, combining behavioural and technical scores without recording the chosen policy would make moderation difficult. The group design consequently treats course, assessment, rubric, question, and student identity as shared context rather than component-specific display labels.

Technical operators support model endpoints, databases, uploaded files, and external dependencies. Their needs include recoverable jobs, protected configuration, clear failure records, and the ability to distinguish a service outage from a valid empty response. The prototype reports a real primary-model failure that was handled through fallback [1], but it does not provide an institutional service-level study. Operational requirements in this report are therefore separated from demonstrated operational performance.

The stakeholder needs are inferred from the documented assessment workflows and the component research questions. The analytics study includes two project-affiliated raters, while formal lecturer acceptance testing remains planned for the grading workflow [1], [4]. These sources provide an initial basis for design; they do not justify a claim that all stakeholder requirements have been independently elicited or confirmed.

1.3 Alignment with the Sustainable Development Goals

The project's primary intended alignment is with Sustainable Development Goal 4, Quality Education, within the United Nations 2030 Agenda [7]. Reviewable feedback and targeted practice could support learning when their subject accuracy and usefulness are established. The relevant mechanism is a shorter and more informative feedback cycle: student work is interpreted against criteria, uncertainty is exposed, and subsequent practice is selected in relation to observed needs. The present studies measure selected technical behaviours rather than that educational outcome.

The project also has a potential relationship with inclusive participation. Handling photographed work and partial recording failures could reduce some practical barriers,

but those same input channels can introduce unequal error rates. Handwriting, English-language fluency, camera quality, and disability may affect what the system can observe. A design that renormalises unavailable signals avoids one direct penalty mechanism, yet cannot establish equitable treatment across groups without authentic data and subgroup analysis [3].

Goal 9, Industry, Innovation and Infrastructure, provides a secondary context for the modular prototype and its reusable interfaces [7]. The contribution is the organisation of different assessment services around shared context and review, rather than a demonstrated change to institutional infrastructure. Sustainability requires maintainable components, documented versions, and a realistic operating model. A prototype dependent on temporary model hosting cannot be treated as durable infrastructure solely because a demonstration succeeds.

Potential negative effects also belong in the evaluation. Incorrect marks, misplaced confidence in machine feedback, inaccessible review procedures, and unnecessary retention of recordings could undermine educational inclusion. Energy and storage demands can grow with video processing and model inference. Chapter 7 therefore frames SDG contribution as a set of outcomes to measure, including review effort, feedback usefulness, equitable completion, and resource consumption, instead of claiming impact from the project's existence.

1.4 Purpose and scope of the common report

This common report explains how the four components form one assessment-support system. It focuses on the shared problem, division of responsibilities, architecture, interfaces, operational dependencies, overall evaluation, and future adoption. Component-specific algorithms and detailed experiments remain attributable to their respective research reports [1]–[4]. Their results are synthesised here where they explain the behaviour or limitations of the complete platform.

The term integration is used at three distinct levels. Architectural integration means that responsibilities and information flows fit together. Implementation integration means that reported code, routes, stored records, or client calls connect those responsibilities. Evaluated integration means that an executed scenario demonstrates

the connection with recorded outcomes. The available material supports these levels unevenly: the handwritten workflow has reported system execution, the analytics thesis describes source-level integration, and the diagram and viva evidence is more limited at complete-system level. For the viva component, this limitation concerns complete-system integration rather than the absence of component-level evaluation: the individual study reports missingness, sensitivity and correlated-degradation experiments, together with automated tests intended to verify score invariance and scoring-state behaviour [3].

The common report does not collapse those levels into one statement of end-to-end success. Results retain their dataset sizes, measurement units, and evaluation conditions. No average accuracy is computed across object detection, segmentation, scoring disturbance, and recommendation quality because these metrics measure different events. Nor is a new human-participant study implied by combining the four documents.

The eight chapters follow the common-report structure. Chapters 1 and 2 establish context and the shared research problem. Chapters 3 to 5 describe the solution, integration, and execution. Chapter 6 evaluates the evidence and objective achievement. Chapter 7 considers sustainability and responsible adoption, and Chapter 8 concludes with prioritised future directions. Appendices provide a proposed integration contract, contribution record, and a test protocol for completing the outstanding group validation.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The overall problem is the absence, in the examined project setting, of a sufficiently traceable workflow that converts heterogeneous computing-assessment evidence into reviewable outcomes and suitable follow-up practice while retaining uncertainty and operational failures. Handwritten text, diagrams, and oral responses require different interpretation methods. Their outputs must still refer to the same educational context and preserve enough detail for an examiner to challenge an incorrect result.

The problem has both assessment and information-management dimensions. Assessment requires a defensible relationship between observed work and the criterion being judged. Information management requires that the right work, question, rubric version, and student remain associated throughout processing. If either dimension fails, a valid-looking final number can be wrong. Consequently, reliable integration is not achieved merely by placing four services behind a common interface.

One example is a diagram-only page. Text recognition may correctly return no useful text, but that does not establish that the student supplied no answer. Another example is a viva recording with an unavailable acoustic feature. Assigning zero to that feature would encode a technical failure as observed poor delivery. A third example is a recommendation derived from a grading placeholder. If the placeholder is interpreted as an approved mark, a service outage becomes a learning weakness.

The research problem therefore requires evidence states as well as numerical values. A result should indicate what was observed, how it was processed, what remains uncertain, and whether it is suitable for downstream use. Gradex investigates component mechanisms that support this requirement, including answer-source labels, structured diagram evidence, modality availability, and traceable topic analytics [1]–[4]. The common challenge is to make those mechanisms coherent across component boundaries.

2.2 Root causes and overall research gap

The first root cause is representation mismatch. A short-answer grader consumes text, a diagram evaluator consumes a graph reconstructed from an image, and a viva scorer consumes temporal measurements. A generic language-model call cannot remove the need to establish those representations. Optical character recognition may produce plausible but incorrect words; line detection may connect nearby but unrelated shapes; and a transcript may omit domain terminology. Each transformation requires independent inspection and evaluation.

The second cause is inconsistent treatment of unavailable evidence. The handwritten prototype uses a flagged emergency response when model services fail, whereas the viva design uses explicit incomplete and pending states [1], [3]. Both approaches make failure visible locally, but they are not numerically interchangeable. The group interface must preserve why a value exists before analytics or export interprets it. A zero is an academic score only when it has the appropriate evidential and review status.

The third cause is the separation between feedback and action. A grading result identifies awarded and missed criteria, but a useful practice recommendation additionally depends on topic labels, cognitive demand, bank coverage, and suitability. The analytics study shows that a more elaborate ranking formula does not necessarily outperform weakness alone [4]. Thus, closing the workflow requires both data integration and an empirical check that the proposed action adds value.

The literature described by the component studies motivates several building blocks. Short-answer grading research concerns how textual responses are scored [5]. Retrieval-augmented generation supplies external context to language models [8]. Low-rank adaptation permits selected parameter updates without retraining every model weight [9]. Multimodal research organises representation, alignment, and fusion problems [10]. The revised Bloom taxonomy supplies a vocabulary for cognitive processes [11]. None of these foundations, by itself, validates Gradex's complete academic workflow.

The overarching gap is therefore an integration and evaluation gap. The project requires an explicit account of how heterogeneous evidence is transformed, how

component errors propagate, which results can be combined, and what each experiment establishes. Its contribution should be assessed through traceability, failure handling, reproducibility, and educational validation, rather than a broad claim that every underlying technique is novel.

2.3 Subproblem and component mapping

The group divides the problem into four interdependent subproblems, summarised in Table 2.1. Component 1 converts photographed or scanned written answers into reviewable per-question grading proposals. Component 2 interprets handwritten entity relationship diagrams as objects, labels, and connections. Component 3 protects a behavioural viva score against inappropriate missing-value treatment. Component 4 transforms graded evidence into interpretable analytics and candidate practice activities.

Table 2.1 Four subproblems and component responsibilities

Component	Student ID	Subproblem	Main output
C1 Handwritten grading	IT22239198	Recognition and rubric-aligned interpretation of written answers	Per-question proposals with transcript and engine evidence
C2 Diagram evaluation	IT22262554	Recovery and interpretation of graphical database structure	Detected objects, connections, reconstructed structure and rubric feedback
C3 Viva score integrity	IT22557056	Scoring permitted delivery evidence when measurements are missing	Behavioural score or status with availability record
C4 Exam analytics	IT22134776	Turning observed achievement into reviewable practice selection	Topic and Bloom summaries, weakness profiles and ranked candidates

These responsibilities avoid treating the platform as four interchangeable graders. C4 primarily consumes assessment evidence and does not replace the upstream marking procedures. C3's behavioural construct is distinct from academic correctness. C2 remains necessary even when a handwritten page contains labels readable by C1, because relationships between shapes carry meaning that a text transcript does not preserve.

The components are interdependent through context and evidence rather than through a mandatory sequence in which every submission passes through all four. A written assessment may contain text and diagrams; a viva may be a separate assessment event. Shared course and student context permits later reporting, but a common identifier does not justify merging unlike scores. The integration strategy therefore preserves assessment modality and score meaning alongside shared identity.

2.4 Overall aim and group objectives

The overall aim is to develop and evaluate an integrated, human-reviewed assessment-support platform that connects heterogeneous student evidence to traceable feedback and appropriate practice recommendations. The aim encompasses engineering feasibility and evidence quality. It does not presume that automated decisions are more valid than examiner judgement or that the prototype is ready for unattended use.

Table 2.2 translates the aim into group objectives and identifies the evidence needed to judge achievement. Each objective includes a result that can be inspected. This prevents a design description from being counted as complete empirical validation and allows component accomplishments to be recognised without inflating the status of the group system.

Table 2.2 Group objectives and evaluation evidence

ID	Objective	Evidence required
O1	Support written, diagram and viva evidence within a coherent workflow	Reported processing paths and shared context
O2	Preserve identity, rubric scope, provenance and meaningful failure states	Interface records and failure-path tests
O3	Generate reviewable assessment proposals and retain examiner control	Evidence displays, overrides and approval records
O4	Link approved assessment evidence to analytics and practice	Traceable ingestion, aggregation and ranking
O5	Evaluate component behaviour and limitations with appropriate baselines	Reported datasets, metrics and comparisons
O6	Establish complete-system readiness for an authentic supervised pilot	Controlled group execution, human validation and operational checks

O1 to O4 address the product's connected behaviour. O5 addresses the research obligation to test what the mechanisms actually achieve. O6 requires evidence beyond a successful demonstration: identity and status must survive the full workflow, reviewers must understand outputs, and real assessment material must be processed under an approved protocol. Chapter 6 reports partial achievement where the required evidence is incomplete.

The corresponding group questions are whether the four responsibilities can be connected without losing evidence meaning, whether failures remain distinguishable from poor academic performance, and whether derived recommendations provide useful action beyond straightforward weakness matching. These questions connect the individual studies while retaining their different units of analysis.

2.5 Scope assumptions and constraints

The implemented and evaluated scope differs across components. Handwritten grading concerns English short answers in Software Architecture, with synthetic examples rather than authentic student scripts [1]. Diagram work concerns five entity relationship object classes and preliminary object detection, with full structural and grading validation still outstanding [2]. The viva component evaluates deterministic, availability-aware scoring through 20,000 sampled feature vectors, thirteen missingness conditions, ±10% input perturbations, parameter sweeps, a separate correlated-degradation simulation, and reported score-invariance tests [3]. Analytics centres on a DBMS question bank, small classification fixtures, and internally rated synthetic profiles [4].

These domains cannot be treated as one common test population. The group's architecture is intended to support shared course metadata, but transfer between Software Architecture and DBMS is not established by that architecture. Each course requires appropriate rubrics, teaching materials, topic labels, and validation examples. Likewise, the five diagram classes do not imply coverage of every notation, cardinality rule, or participation constraint.

The system assumes that a lecturer supplies a meaningful rubric and reviews the resulting evidence. OCR and inference depend on image quality, available services,

and model configuration. The prototype includes hosted OCR and model routes, while viva processing also depends on recording and feature quality. The analytics output depends on the accuracy and coverage of upstream marks and question-bank labels. None of these dependencies can be eliminated by a shared web client.

The project excludes demonstrated future-examination prediction, autonomous final grading, validated inference of competence from emotion, and measured learning improvement. Commercial feasibility, broad accessibility, institutional security assurance, and sustained production operation remain evaluation tasks. Stating these boundaries enables an honest account of the completed prototype and a practical plan for the work required before institutional adoption.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

Gradex organises assessment support around a shared lecturer-facing client, server interfaces, course and rubric context, and component services. Student work enters an appropriate processing route. Each route produces evidence and a provisional interpretation, which should remain available for review. Approved assessment evidence can then support student and cohort analysis, while practice candidates are presented as suggestions for further teaching or learning.

The central value proposition is continuity of evidence. A lecturer should be able to move from a displayed result to the underlying answer, structure, feature set, or rubric criterion. A recommendation should be explainable through the assessment evidence and course material that produced it. Continuity is valuable even when the automated interpretation is imperfect, because the reviewer can locate and correct the source of an error rather than only changing a final total.

The system does not require one universal scoring engine. Written-answer evaluation is language-model-based, diagram evaluation combines vision and structural interpretation, and viva Stage-1 uses deterministic family aggregation. Analytics uses deterministic numeric calculations alongside model-based semantic interpretation. This mixture is deliberate: different inputs call for different mechanisms, and each mechanism carries its own verification requirements [1]–[4].

At group level, the proposed review boundary links these mechanisms. Provisional output, approved academic evidence, diagnostic information, and generated advice should remain separate categories. The current component accounts document useful parts of this boundary but do not prove one shared approval implementation spanning every route. The integrated architecture therefore distinguishes reported connections from the stronger acceptance contract proposed in Appendix A.

3.2 High level architecture

Figure 3.1 presents the integrated architecture as a synthesis of the component descriptions [1]–[4]. The shared client supports course setup, evidence submission, review, and analytics. The shared server provides access to the processing services.

Handwritten grading and diagram evaluation transform submitted material; viva evaluation processes recorded oral evidence; analytics consumes assessment records and course resources. External OCR and inference providers are separate dependency boundaries.

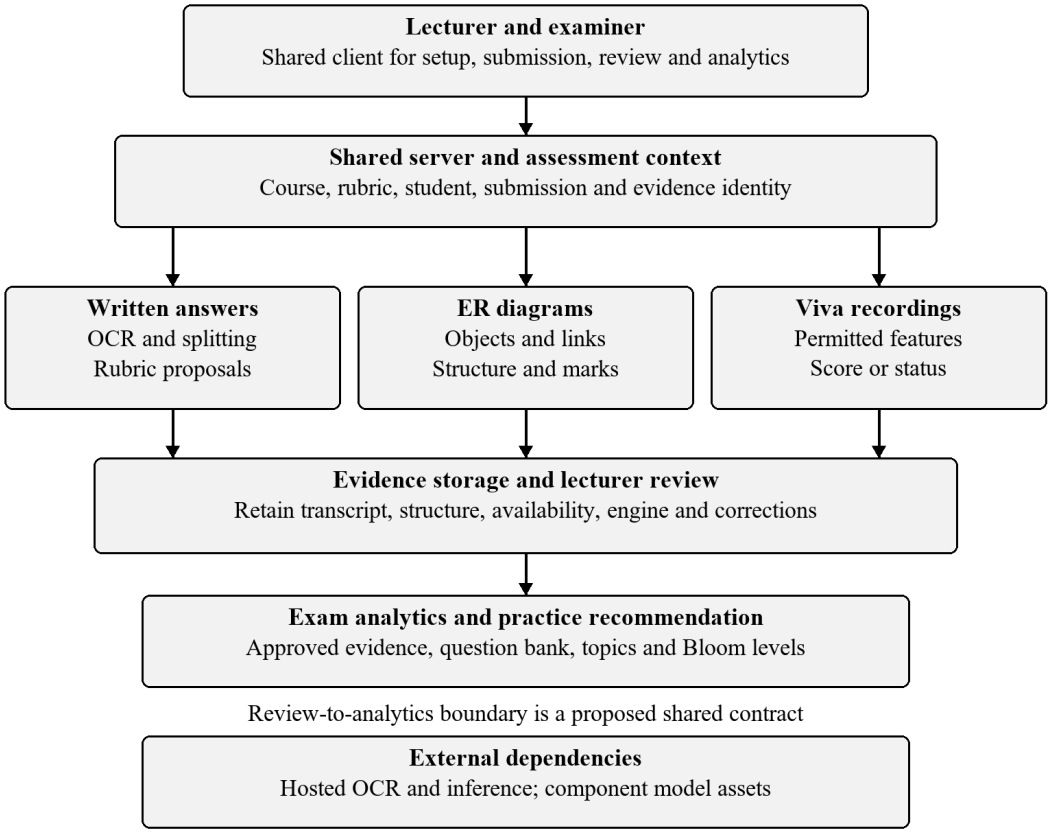


Figure 3.1 Integrated assessment and practice architecture

MongoDB is described as the principal persistence mechanism for course, rubric, submission, progress, and analytics records. The handwritten component additionally maintains a course-scoped ChromaDB index for lecture material [1]. The architecture should not be interpreted as proof that every service shares an identical database schema or security policy. The grading and analytics accounts describe several concrete readers and writers, while exact viva and diagram API contracts are not fully specified in the available reports.

Data flows have different evidential meanings. Submission flows carry raw or extracted work. Evaluation flows carry proposed marks or behavioural scores with provenance. Analytics flows carry normalised assessment evidence and derived

summaries. Advisory flows carry generated explanations or practice questions. These distinctions matter because a failure in an advisory language model should not change deterministic mark totals, and diagnostic video outputs should not silently enter Stage-1 scoring.

The shared client is a presentation and interaction boundary rather than evidence of validated score fusion. Similarly, a common server process is an implementation choice rather than a guarantee of independence between engines. The handwritten thesis describes module-name collisions when multiple Python packages expose an app package [1]. This is an integration risk that requires regression coverage even when the high-level service boundaries appear clean.

3.3 Component 1 summary

The handwritten grading component is attributed to H.I.B. Wickramarathne, IT22239198. It receives photographed or scanned answer scripts, a selected course, a rubric, lecture materials, and roster information. Its purpose is to turn noisy paper-based answers into per-question grading proposals with enough evidence for lecturer review. The documented service is GradingEngine, mounted at /api/grading [1].

The pipeline lightly conditions images, applies an OCR cascade, removes recognition banners and other noise, and maps detected question markers to canonical question numbers. Each question receives an answer-source label. Split means a question-specific bucket was found; full means no markers were available and the whole transcript is used; empty means other markers exist but the selected question has no corresponding answer. These labels make segmentation uncertainty visible rather than concealing it in the prompt.

Retrieval selects course-tagged lecture chunks through MiniLM embeddings and ChromaDB. The query incorporates the topic, question, and rubric criteria. The primary grader is Llama 3.1 8B adapted with LoRA on 1,250 synthetic examples, distributed equally across five answer behaviours. A hosted fallback is used when the primary route fails, followed by a flagged emergency response if neither route is available [1]. The final result includes score proposals, justification, answer-source information, engine identity, and retrieval context.

Roster validation and batch orchestration support the operational workflow. Manual identifiers take priority over folder-derived identifiers and OCR-derived identifiers, and ambiguous cases remain visible. The client allows review, annotation, override, and regrading. Its contribution to the group system is both the written-answer interpretation path and the persistent per-question evidence that can later support analytics.

The evidence limit is substantial. Synthetic segmentation, marker, retrieval, and software tests demonstrate selected behaviours; they do not establish agreement with human markers on authentic scripts. The detailed results also identify unresolved marker errors and an unexpected ordering of training-label means. The component must therefore remain a source of reviewable proposals rather than independently validated marks.

3.4 Component 2 summary

The diagram evaluation component is associated with IT22262554 and the paper led by A. M. Senarathna [2]. It receives an image of a handwritten entity relationship diagram and an optional marking guideline. The aim is to recover enough graphical structure to evaluate criterion-level evidence rather than assessing only the presence of text labels.

YOLOv8l identifies five classes: attributes, entities, primary keys, relationships, and weak entities. Each detected object receives a class, bounding box, confidence, identifier, and crop. OCR is then applied to processed crops, using OCR.space Engine 3, and labels are associated with the corresponding objects. Cached OCR results are identified through a SHA-256 digest of the processed crop. The cache concerns repeated recognition input; it does not validate the correctness of the recovered label.

OpenCV-based edge and line processing identifies potential connections. The reconstruction approach associates line endpoints with nearby detected objects and converts the resulting objects and links into a structured entity relationship representation. A locally hosted language model evaluates this structure against rubric criteria. The grading instruction constrains the model to extracted evidence, requires

one result per criterion, and prevents invented objects or connections from justifying marks [2].

Outputs include annotated images, detected objects, OCR labels, connection records, reconstructed structure, criterion-level marks, and feedback. These intermediate artefacts are particularly valuable for review because a wrong final result may originate in detection, recognition, association, or grading. The analytics thesis describes readers for diagram evaluation and marking records, providing a reported implementation connection to a separate diagram-analysis view [4].

The preliminary experiment trained on 350 images and evaluated five validation images. It reported 88.4% precision, 61.3% mAP@0.5, and 37.1 milliseconds per image for detection [2]. These values concern the detector, not the entire diagram assessment. OCR, connection accuracy, structural reconstruction, and agreement with expert grading remain unmeasured in the paper.

3.5 Component 3 summary

The viva score-integrity component is attributed to B.K.H.M.B.L. Herath, IT22557056. The documented implementation identifies the scoring core as VivaEvaluationEngine version 1.1. Its official behavioural score combines three feature families: visual engagement, using mean CNN engagement and facial confidence; acoustic delivery, using pitch stability, HNR clarity and articulation derived from jitter and shimmer; and transcript fluency, using speech-rate band, hedges, fillers, long pauses and sentence completion. Each input follows a defined eligibility policy and bounded deterministic mapping before family-level aggregation [3].

Within each family, available normalised sub-features are averaged. Families with no available sub-feature are omitted, and the remaining family weights are renormalised. An unavailable feature therefore does not become an observed zero. The documented capture-integrity checks include face-quality assessment, multiple-face detection and consistency between lip motion and transcribed speech. A failed integrity gate or the absence of scorable evidence should produce the configured status outcome rather than

an ordinary low academic score [3]. This policy separates processing failure from measured performance while keeping the calculation auditable.

Stage-1 explicitly excludes emotion labels, raw gaze and blink measures, several acoustic diagnostics, and language-model criterion scores. Some computable signals are held out for further validation. The architectural boundary is as important as the formula: additional upstream outputs must not alter an official score merely because they exist. The thesis reports invariance tests intended to protect this separation [3].

Mode A returns the 100-point behavioural score. Mode B combines that score with an examiner-entered technical-accuracy score T, defined on a 0–10 scale, using equal weighting: 50% behavioural score and 50% normalised technical accuracy. The combined result remains a draft until the technical score is supplied and the examiner publishes the result. Where the required technical score is absent, the result remains pending rather than being treated as a completed assessment [3]. Live examiner assistance remains advisory; live and uploaded-recording routes are described as converging on the same scoring pathway.

The reported study uses 20,000 synthetic feature vectors across thirteen missingness conditions. Renormalisation reduced mean absolute disturbance from 9.12 to 3.04 points and grade-band changes from 65.8% to 29.5%; under correlated degradation, disturbance was 2.82 versus 9.09 points and grade-band changes were 26.3% versus 62.6%. Separate sensitivity experiments reported a maximum 1.52-point movement under ±10% perturbation and mean score changes no greater than 1.15 in constant sweeps. The implementation inventory reports 189 passing tests across 16 files, although the test files and raw simulation artefacts were not supplied for independent reproduction. These results concern simulated score behaviour, not real-viva validity [3].

3.6 Component 4 summary

The analytics component is attributed to Liyanage Supun Buddhika Hemarathne, IT22134776. Its registered title includes exam question analytics and prediction, while the evaluated AdaptiveExamAnalyticsEngine performs assessment analysis and

weakness-aligned practice recommendation [4]. It does not implement or validate a forecast of future examination questions.

The engine receives course metadata, rubrics, graded submissions, and question-bank records. A local ModernBERT classifier assigns one of six revised Bloom categories. Topic assignment and taxonomy handling associate questions with course concepts. Deterministic calculations aggregate awarded and maximum marks across students, questions, topics, and cognitive levels. These outputs support an observed weakness profile, not a validated estimate of latent mastery.

Candidate practice questions are ranked using a five-signal heuristic: topic weakness, lecture coverage, tutorial evidence, recent-exam relevance, and a Bloom-level gap. Optional language-model generation produces follow-up items or teaching actions after the numerical and ranking steps. Generated suggestions are distinct from direct calculations, and their pedagogical correctness requires review.

The thesis describes shared submission, course, and rubric readers, persistent analytics collections, diagram-analysis inputs, and matching client calls. It also attributes the entire group client-side interface design to the analytics author [4]. The handwritten account separately describes development of its grading workflow [1]. The contribution record in Chapter 4 preserves these reported roles and leaves the exact shared implementation split for team confirmation.

Evaluation uses an eleven-question classifier fixture and ten synthetic weakness profiles with two project-affiliated raters. The classifier matched all fixture labels, while the default ranker matched weakness-only precision and had slightly lower NDCG point estimates. The study therefore supports the existence and inspection of an evidence-to-action pipeline without demonstrating added ranking value from every signal or improved learning outcomes.

3.7 End to end user and data flow

The proposed complete workflow begins with a lecturer selecting the course, defining the assessment and rubric, and checking roster identity. Lecture resources and question-bank records provide context for grading and follow-up practice. A submission is then routed according to its evidence type. Written answers enter

recognition and segmentation, diagrams enter structural extraction, and viva recordings enter integrity checks and feature processing.

Each route returns both an interpretation and the evidence needed to review it. The written route exposes transcript slices, criterion proposals, retrieval context, and the answering engine. The diagram route exposes objects, labels, connections, and reconstructed structure. The viva route exposes permitted feature and family contributions, unavailable inputs, integrity-gate outcomes, scoring mode and completeness status, allowing a reviewer to distinguish measured delivery performance from incomplete evidence. Review can identify whether a surprising result arose from the student's work or from a processing stage.

Figure 3.2 shows the review boundary and the intended progression to analytics. It is a workflow specification derived from the component descriptions, not a claim that this exact four-component scenario has already been executed. The decisive transition is approval of usable evidence. A processing failure, unreviewed emergency placeholder, or incomplete viva result should not enter academic analytics as an ordinary low score.

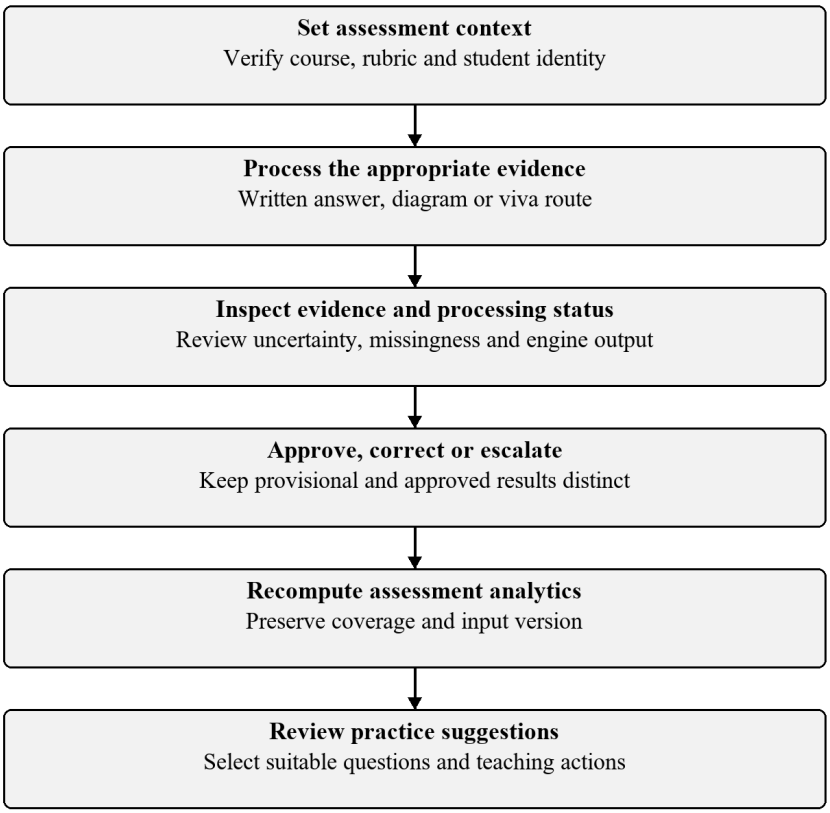


Figure 3.2 Review boundary between processing and downstream analytics

After appropriate review, assessment records can support topic and Bloom summaries. The lecturer examines coverage and uncertainty before interpreting low achievement. Candidate practice is ranked from the available bank, and generated material is reviewed before delivery. A correction to an upstream result should trigger corresponding reanalysis so that later advice does not remain based on superseded evidence.

This workflow accommodates different assessment events without forcing every student through every modality. It also keeps the student-facing output understandable: approved feedback, behavioural observations, and practice suggestions have different meanings. The next integration study should demonstrate these distinctions through one recorded scenario with stable identifiers, explicit state transitions, and observable recovery paths.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The reported integration strategy is modular service composition around shared user and assessment context. The handwritten engine is mounted as a FastAPI sub-application, while the analytics engine exposes student, lecturer, and course-taxonomy routes [1], [4]. MongoDB records and client adapters connect component functions. Diagram and viva processing retain their own algorithms and evidence structures. This division allows specialised processing without requiring every component to adopt the same model or numeric scale.

Integration should proceed from shared identifiers to usable evidence, then from evidence to derived outputs. First, a course, assessment, rubric, student, and submission must refer to the same entities across services. Second, processing results must retain source and status. Third, the client must present the result for review. Finally, analytics should consume only records whose intended meaning is supported. The component reports demonstrate parts of this sequence but do not provide one versioned release record for the entire platform.

The shared-server arrangement introduces a specific implementation concern. Several engines use app as a top-level Python package. The handwritten thesis describes importing its application first, keeping direct references, and restoring module-cache entries around later imports [1]. This workaround can support a prototype, but the integrated test plan must exercise import order, database helpers, and route registration together. A component's isolated tests cannot establish that another engine has not replaced a shared module reference.

The strongest current integration evidence concerns the handwritten client-to-server workflow and the analytics source interfaces. The diagram paper documents an image-processing endpoint but not its exact external contract. The viva thesis describes a canonical feature boundary but does not supply endpoint definitions or storage schemas [2], [3]. Appendix A therefore states a proposed common contract rather than presenting invented endpoints as implemented facts.

4.2 Interfaces data contracts and shared models

Table 4.1 distinguishes reported interface elements from the proposed common interpretation needed for dependable integration. A shared contract should carry stable identity, rubric scope, evidence references, processing status, engine version, and review state. It should also distinguish original outputs from later corrections. These fields are logical requirements; exact field names and serialization should be reconciled with the actual component schemas before release.

Table 4.1 Interface evidence and common contract requirements

Boundary	Reported element	Required common interpretation
Client to grading	/api/grading; roster and rubric context [1]	Verified student and assessment association
Grading to storage	Transcript, per-question result, engine and session identity [1]	Provisional score kept separate from approval
Diagram processing	Image plus optional guideline; structured result [2]	Object and connection provenance retained
Viva scoring	Versioned Stage-1 features, eligibility and availability [3]	Retain score, available-family set, integrity-gate outcome, scoring mode and policy version together
Analytics ingestion	Submission, course, rubric and diagram readers [4]	Input version and eligibility for analysis recorded
Analytics to client	Student, lecturer and taxonomy interfaces [4]	Calculated values separated from generated advice

Identifier consistency is essential. The same student identifier should not be independently guessed at every stage. The grading roster process explicitly prioritises a manual override, then a valid folder identifier, then OCR [1]. Downstream components should consume the resolved identity with its provenance. A duplicate or unreadable identity should remain unresolved until reviewed; a numeric score must not make an ambiguous student association appear complete.

Rubric identity also requires versioning. A criterion's maximum mark determines the denominator of later percentages. If a rubric is edited after processing, previously calculated totals may no longer be comparable. A defensible contract therefore associates results with the rubric version used for evaluation and preserves regrading as

a new result or auditable revision. This is a proposed group-level requirement, not an assertion that every component already implements historical rubric versions.

Status semantics need explicit reconciliation. The written engine reports a flagged zero when neither model route can answer. Its empty-answer route also returns zero, but for a different reason [1]. The viva design can return INCOMPLETE or leave Mode B pending [3]. These outcomes should not be normalised into the same academic score. Appendix A proposes separating processing state, evidence state, review state, and numeric value so that consumers can make a justified eligibility decision.

Scale metadata prevents another class of integration error. Written and diagram marks belong to rubric maxima, the viva behavioural score uses a 100-point scale, and external technical accuracy T is defined on a 0-to-10 scale before optional fusion [3]. A common dashboard may display all of them, but it must not average them without a declared assessment policy. Practice ranking scores are heuristic ordering values and should never be presented as academic marks.

4.3 Shared nonfunctional requirements

Reliability means preserving evidence and making degraded operation visible. The handwritten engine catches failures per script and can switch inference routes, but its in-process batch jobs can be lost on restart [1]. Availability-aware aggregation permits scoring when sufficient permitted evidence remains and the integrity gates pass. When these conditions are not met, the component can return an explicit incomplete state rather than treating missing evidence as poor delivery [3]. Reliability should therefore be measured as correct completion or correct escalation, not simply the proportion of requests returning a number.

Performance is currently supported by limited and non-comparable observations. The diagram detector's 37.1 milliseconds per image excludes OCR, structural reconstruction, grading, storage, and reviewer interaction [2]. The handwritten failover event lacks recorded retry timings, and the group reports do not provide complete-system throughput [1]. A future performance evaluation should separate queue delay, extraction, inference, persistence, and review time, then report distributions under a stated hardware and workload configuration.

Usability requires clear evidence displays. The lecturer should understand which model produced a proposal, which parts of the submission were used, and what an unavailable input means. The reported grading interface exposes engine labels, context, and overrides [1]. Comparable review affordances are needed for diagram connections and viva availability. Accessibility should be evaluated with actual users and alternative review routes rather than inferred from use of a web interface.

Privacy and security requirements follow from the sensitivity of student identifiers, scripts, marks, and recordings. Course-scoped retrieval helps prevent irrelevant context from entering a grading request, but is not a complete access-control policy. Institutional deployment should verify authorization, secure transport, upload validation, retention, deletion, and audit access. The report treats these as acceptance requirements and does not claim a completed security audit or legal certification.

Maintainability depends on explicit boundaries and reproducible versions. A change to topic taxonomy can invalidate cached classification or recommendations. A change to the viva feature ledger can alter score meaning. A change to the grading model can affect results even when the API is unchanged. Model checksums, schema versions, rubric versions, and scoring-policy versions should therefore be recorded together for each release.

Scalability requires attention to shared bottlenecks. Hosted OCR quotas, a single inference notebook, database contention, and video processing can affect different routes unevenly. The architectural response is to isolate long-running work and measure resource use before selecting a production topology. The current prototype offers a basis for these measurements but not evidence that it can support an institutional examination period at scale.

4.4 Integrated workflow and exception handling

Normal processing should retain the original submission, the selected rubric, intermediate evidence, provisional output, and reviewer decision. Reanalysis should be explicit so that an amended transcript or corrected diagram connection can be traced to the resulting mark. The final approved result is only one part of the record. Without

earlier evidence, a later moderation request cannot determine whether the issue arose in recognition, interpretation, or review.

Table 4.2 maps important failure paths to their group-level implications. Some behaviours are reported implementations, while downstream handling is an acceptance requirement. The distinction matters because component-level safeguards can be lost during integration. A clear flag in one service provides little protection if the next service reads only the numeric field.

Table 4.2 Failure paths and downstream handling

Condition	Reported component behaviour	Group-level handling required
No usable OCR text	Script may be skipped or flagged [1]	Preserve upload and reason; do not infer no academic attempt
Missing question bucket	Empty-source zero with reason [1]	Review segmentation before finalising the mark
Both grading engines fail	Flagged emergency zero [1]	Exclude from academic totals until resolved
Diagram connection uncertainty	Nearest-object association can fail [2]	Show structure for review; avoid unsupported criterion credit
Missing viva family	Renormalise remaining families [3]	Show coverage and retain score meaning
Failed viva integrity gate	INCOMPLETE where specified [3]	Escalate for examiner review rather than assign failure
Missing technical score	Mode B remains pending [3]	Do not publish fused result
Missing Bloom model	Active classifier path fails [4]	Surface failure; do not claim documented fallback occurred
Unclassified topic	Retained overall, excluded from named-topic conclusions [4]	Show coverage and request topic review

Retries should preserve identity and avoid duplicate academic effects. The handwritten router retries transient service failures, while batch progress records the current stage [1]. The group acceptance contract should verify that a retried submission does not create two approved marks or duplicate analytics evidence. This idempotency requirement is particularly important when a client loses a response after the server has already persisted a result.

Degraded model routing also changes provenance. A batch may contain proposals from both the adapted local model and the hosted fallback. Completion therefore does not establish consistent model behaviour across all questions. The review interface should retain per-question engine labels and support comparison when differences matter. A later study should evaluate the primary and fallback models on identical approved inputs instead of treating availability as accuracy.

Corrections must propagate forward. If a lecturer changes an answer boundary, mark, or topic label, previously computed weaknesses and recommendations may become stale. A common release should define which changes invalidate derived records and how the client communicates that reanalysis is required. The analytics source describes versioned outputs and taxonomy-sensitive caching [4], but a full correction-to-recommendation regression test remains necessary.

4.5 Team contribution and dependency matrix

Table 4.3 consolidates the contributions stated in the four reports. It is an attribution record for the common thesis and should be confirmed by all members before signing Appendix B. The diagram paper is co-authored; its association with IT22262554 follows the supplied file identification, while the full student name remains for completion. Shared client design and grading-workflow implementation are recorded separately because the source accounts describe overlapping levels of responsibility.

Table 4.3 Team contributions and integration dependencies

Student	Research and development contribution	Main integration dependency
H.I.B. Wickramarathne, IT22239198	Handwritten grading, synthetic corpus, adapted model, retrieval, roster, batch and grading workflow [1]	Course and rubric context; review client; stored evidence for analytics
IT22262554	ER detector, OCR association, connection reconstruction and rubric evaluation described in the diagram paper [2]	Image input and guideline; diagram evidence readers
B.K.H.M.B.L. Herath, IT22557056	Stage-1 feature eligibility boundary; deterministic family aggregation and weight renormalisation; Mode A/Mode B scoring policy; and reported robustness and invariance analysis [3]	Valid feature extraction; recording gates; external technical score

Student	Research and development contribution	Main integration dependency
Liyanage Supun Buddhika Hemarathne, IT22134776	Analytics engine, classification and ranking evaluation, complete group client-side interface design [4]	Approved grading data; course taxonomy; curated question bank

The dependencies explain why individual completion is not equivalent to group completion. Analytics cannot establish the validity of upstream grades merely by aggregating them correctly. Diagram grading cannot recover a missed object through a stronger language-model prompt without departing from its evidence constraint. Viva renormalisation cannot repair invalid feature extraction. The group system must therefore preserve both component ownership and responsibility for shared contracts.

Shared outputs include the common architecture, consistent result interpretation, integration test specification, contribution declaration, and final report. The reports do not supply a consolidated record of commits, reviews, meetings, or task assignments proving every shared activity. Chapter 5 therefore distinguishes documented contribution claims from process evidence that must still be assembled. This protects the common report from attributing undocumented activities or implying that all members performed identical work.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The reported platform uses a React and TypeScript client with Python services, FastAPI interfaces, and MongoDB persistence [1], [4]. The shared technologies enable a common user workflow while the specialised processing remains component-specific. Table 5.1 summarises their roles. The stack is reported as a prototype configuration; it is not a claim that every component was tested in one reproducible environment or deployed with a common dependency lockfile.

Table 5.1 Shared and component-specific technologies

Layer	Reported technologies	Purpose and boundary
Client	React, TypeScript, Vite	Lecturer workflows, review and analytics
Services	Python, FastAPI, Uvicorn, Pydantic	API handling and component orchestration
Persistence	MongoDB; ChromaDB for grading retrieval	Assessment records and course-filtered context
Written evidence	OpenCV, OCR.space, MiniLM	Image preparation, recognition and retrieval
Written grading	Llama 3.1 8B, LoRA, Unsloth; Groq fallback	Adapted primary inference and degraded routing
Diagram evidence	YOLOv8l, OpenCV, OCR.space, local LLM	Detection, labels, connections and rubric interpretation
Viva evidence	OpenCV, FFmpeg, MediaPipe Face Landmarker, EfficientNet-B0, Whisper, Praat/Parselmouth and librosa	Media and feature extraction for visual engagement, acoustic delivery and transcript fluency. The EfficientNet-B0 engagement model supplies an official scoring input; emotion-analysis models, speech-emotion outputs and LLM services provide supporting or advisory functions separate from the official behavioural score [3].
Analytics	ModernBERT; optional MiniLM and Ollama	Cognitive classification, similarity and advice generation

The handwritten model is described as hosted through a Colab notebook and ngrok, with external OCR and hosted fallback inference [1]. These choices fit a resource-constrained prototype but introduce changing endpoints, quotas, and session lifetime. The diagram route also depends on external OCR, despite local language-model grading [2]. A deployment diagram must therefore show what leaves the institution's runtime rather than describing the entire platform as local.

The viva account reports Python 3.10.11 for the scoring paper, while the handwritten account describes Python 3.12 on its server [1], [3]. These are component-specific environment statements. They should not be silently merged into one verified runtime. Integration requires either compatible isolated services or a tested common environment, with model and library versions recorded at the time of execution.

The analytics thesis lists API and model dependencies and describes a local ModernBERT checkpoint, cache identity, and optional model-generation routes [4]. Its source observations are more detailed than its historical experiment record. A release manifest should therefore distinguish the evaluated research version from a later implementation snapshot, especially when topic matching, cache behaviour, or tie handling has changed.

5.2 Development and integration process

The component accounts describe an iterative design-and-evaluate process. The handwritten engine uses independently inspectable stages and tests deterministic splitting, identity, and model-health logic. The diagram paper separates detection from later structural and grading stages. Viva evaluation isolates the scoring policy from upstream diagnostic outputs. Analytics separates deterministic calculations from semantic interpretation and generation [1]–[4]. These choices support local debugging and evidence-specific evaluation.

At group level, development should be organised around a stable integration scenario. The first milestone is a consistent course, rubric, and submission record. The next is successful processing with inspectable evidence. The third is reviewer approval or escalation. The final milestone is a traceable analytics output and practice

recommendation. This sequence is a proposed completion process; no sprint schedule or continuous-integration history is claimed without records.

Version control should capture both code and evaluation artefacts. The analytics thesis identifies a checked-out revision reference and a later local-file inspection, but explicitly distinguishes that reference from proof of an unchanged working tree [4]. That distinction applies to the whole project. A tag alone is insufficient if model weights, training data, or runtime configuration are outside the repository and undocumented.

Interface review should precede a combined test. Each component owner should confirm identifier names, required inputs, score ranges, missing-value representation, and error responses. The group should then test deliberately malformed and incomplete requests. This is particularly important where one engine returns a flagged number and another returns no number. A contract review can expose such disagreements before they become misleading dashboard data.

Regression should follow changes that alter downstream meaning. Examples include a corrected marker parser, a new diagram object class, a changed normaliser, or a revised taxonomy. The affected tests should include the consumer as well as the producer. A locally correct change can still break integration if a client assumes an old field name or analytics caches an old interpretation.

5.3 Deployment and operational configuration

The handwritten prototype is described as running the shared server, MongoDB, and local vector index on one machine, with primary inference in Colab and external OCR and fallback services [1]. The operator configures the current evaluation endpoint and service credentials. A cached health verdict helps avoid repeated calls to a dead primary route, but the reported incident shows that a live health endpoint can coexist with failing evaluation requests.

The component's reported defaults include a 20-second connect timeout, 120-second read timeout, up to three retries, a 60-second health cache, and OCR parallelism of three [1]. These are configuration facts from the component account, not a recommendation that they are optimal. Complete-system latency depends on the

number of pages, questions, retries, and active jobs. A deployment study should measure the effect of these settings rather than infer it from individual timeouts.

The analytics service reads local classification assets and database records, with optional generation services configured separately [4]. Its active classification route depends on model availability. Numeric advice fallbacks do not imply that every semantic stage has a fallback. Operational checks should therefore verify model presence, tokenizer and weight compatibility, database indexes, and generation status independently.

Before an institutional pilot, the deployment package should include a release manifest, environment configuration with secrets excluded, model checksums, database migration or index instructions, and a recovery procedure. A controlled restore should verify that approved results and evidence associations survive recovery. Background jobs should have a documented restart policy. The current reports do not establish completed backup, restore, disaster-recovery, or long-duration monitoring tests.

Operational logs should record the stage, reason for failure, selected engine, version, and relevant record identifiers without unnecessarily duplicating raw recordings or scripts. Review and correction events should remain auditable. The aim is to reconstruct an assessment decision while limiting unnecessary copies of personal evidence. Retention periods and access roles require institutional decisions before authentic-data collection.

5.4 Evidence of team execution

The strongest documentary evidence of execution consists of the four component accounts and their reported artefacts. The handwritten thesis reports an implemented pipeline, 21 passing tests across three modules, manual synthetic workflows, and one observed fallback event [1]. The diagram paper reports detector training and preliminary validation [2]. The individual viva thesis reports a 20,000-vector missingness study, input-perturbation and parameter-sensitivity experiments, correlated-degradation testing, score-invariance checks, and 189 passing test functions

across 16 files [3]. The analytics thesis records a repository inspection and selected offline checks [4].

These evidence types should remain distinguishable. A reported passing test is not a freshly reproduced result in the common report. A source-level API match is not a browser execution. A trained model is not validated against independent human marks. The group account uses the original reports as evidence for what each member documented, while avoiding claims that repositories, logs, or raw experimental files were independently rerun during synthesis.

No consolidated MS Planner export, Teams history, meeting record, research diary, or group release log accompanies the component evidence. The report therefore does not invent sprint dates, completed reviews, task durations, or attendance. Appendix B provides a concise declaration that the team can confirm, and Appendix D identifies the artefacts needed to support a stronger execution record.

The source documents also contain editorial inconsistencies. The handwritten abstract characterises a 14-case result as malformed grading-response recovery, while its detailed chapter reports marker-shape normalisation. Its detailed segmentation table shows four zero-baseline conditions, whereas another summary gives a different count [1]. The common results use the detailed tables, retain neutral metric labels where definitions are incomplete, and avoid resolving unsupported claims by guesswork.

The analytics source includes missing or corrupted sections and identifies differences between active code and a described fallback policy [4]. The viva account states that its repository and raw simulation artefacts were not available for independent reproduction [3]. These limits define the current evidence record. Closing them is part of completing research verification, not simply improving the presentation of the thesis.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The group evaluation separates processing correctness, score validity, operational integration, and educational usefulness. Processing correctness asks whether each stage transforms the stated input as intended. Score validity asks whether the result agrees with a defensible assessment construct and expert rubric interpretation. Operational integration asks whether identity, provenance, state, and correction survive the complete workflow. Educational usefulness asks whether the feedback or recommended action benefits the intended users.

Table 6.1 maps these dimensions to evidence and acceptance conditions. Conditions marked prospective are proposed for the next controlled group study and are not reported as passed. This separation prevents the detector's precision or a software-test count from substituting for academic validation.

Table 6.1 Group evaluation framework

Dimension	Current evidence	Prospective acceptance condition
Processing	Synthetic splitting and retrieval; preliminary detection; fixture classification	Correct attribution and bounded outputs on independently prepared cases
Score integrity	Reported missingness, sensitivity, correlated-degradation and invariance tests	Missingness and excluded signals retain their specified semantics
Academic validity	No authentic group marker-agreement study	Independent rubric ratings with reliability and agreement analysis
Integration	Reported grading workflow and analytics source connections	One traceable scenario across all relevant services, including failures
Usefulness	Internal recommendation ratings	Independent lecturer review and measurable workflow benefit
Operation	One reported failover; no group load study	Defined recovery, access-control, and workload tests

The datasets are intentionally reported separately. The handwritten tests use synthetic scripts, marker forms, and retrieval queries. The diagram experiment uses 350 training images and five validation images. Viva scoring uses 20,000 sampled feature vectors. Analytics uses an eleven-question classifier fixture and 100 candidate rows across ten

synthetic profiles [1]–[4]. None of these sample sizes represents the number of authentic students evaluated by the complete platform.

Baselines are also component-specific. The written splitter is compared with a minimal Arabic-digit baseline; the viva policy is compared with zero-imputation; and the analytics ranker is compared with random and weakness-only orderings. These comparisons answer useful local questions but do not establish superiority over all available assessment systems. No direct competing complete-platform experiment is reported.

The evaluation should retain denominators and uncertainty. A success on five diagram images or eleven explicit questions has limited generalisability. A high count of synthetic feature vectors gives precise coverage of the chosen sampler without validating the human construct. A bootstrap interval across ten profiles expresses variability within that pilot design rather than representing all courses or student populations.

6.2 Integration and end to end testing

The handwritten thesis reports that the mounted service and React workflow completed session setup, roster validation, recognition, segmentation, retrieval, grading, and review using synthetic scripts [1]. It also reports a batch in which a successful health probe was followed by HTTP 500 evaluation failures. Retries and cache invalidation redirected later questions to Groq, allowing the batch to complete with engine labels. This is evidence of one observed recovery path, not an availability estimate.

The same account reports 21 passing unit tests: six splitter tests, nine roster tests, and six health-logic tests. Wrapper parsing and a per-question smoke script provide additional assets, but they should not be added to the 21 as if all were counted using the same unit [1]. The tests do not establish the accuracy of the fine-tuned grader, live OCR recognition, or the complete browser interface.

The viva thesis reports 189 passing test functions in 16 files, including 15 scoring-core targets and 174 surrounding tests, with a reported re-verification date of 1 September 2026 [3]. The reported checks cover invariance to diagnostic and LLM-generated

scores, grade boundaries, incomplete states, capture-integrity gates and Mode B behaviour. Because test files, run logs and raw simulation artefacts were not supplied for this synthesis, these remain reported component-level findings rather than independently reproduced results.

The analytics thesis describes 41 test files and five existing pure test functions successfully invoked in its supplementary review [4]. The full suite was not run in that review, and the isolated checks excluded database fixtures, live services, browser workflows, and model inference. Source connections to the shared client and diagram readers are therefore implementation evidence with a narrower scope than a measured full-platform run.

Table 6.2 summarises the group integration status. Appendix C expands it into a reproducible test protocol. The present reports do not establish one controlled run spanning written and diagram evidence, viva status handling, reviewer approval, analytics refresh, and practice selection. That missing experiment is the principal boundary on an overall integration claim.

Table 6.2 Integration evidence status

Scenario	Evidence status	What remains unestablished
Written script to review	Reported synthetic manual execution [1]	Authentic marker agreement and formal usability
Primary model outage	One reported fallback event [1]	Long-run reliability and latency distribution
Diagram image to final expert grade	Detector evaluated; later stages described [2]	Structural and grading accuracy
Viva inputs to protected score	Reported robustness simulations and 189-function test suite [3]	Independent reproduction and real-viva validity
Graded records to analytics UI	Source interfaces and selected offline checks [4]	Live complete-system execution
Correction to updated practice	Proposed group acceptance scenario	Recorded execution and stale-output prevention

6.3 Overall results

The results support a modular prototype with useful engineering mechanisms and uneven empirical maturity. They should be interpreted by stage and dataset rather than reduced to a single platform accuracy. The following subsections retain the measurements and limitations that matter most to the integrated workflow.

6.3.1 Handwritten recognition and retrieval results

Table 6.3 reproduces the detailed splitter comparison from the handwritten thesis [1]. Each condition contains 200 synthetic scripts. The source calls the two measurements a first and a stricter second score without providing sufficiently precise definitions in the extracted text. They are therefore retained as Score 1 and Score 2, rather than relabelled as precision, recall, or exact-match accuracy. This preserves the result without inventing a metric definition.

Table 6.3 Reported segmentation scores in percent

Condition	Proposed Score 1	Proposed Score 2	Baseline Score 1	Baseline Score 2
Arabic clean	100	100	81.6	71.0
Enumerated	100	100	0	0
Roman	100	100	0	0
Roman glyph substitution	83.0	68.9	0	0
Arabic with banners	100	100	82.0	69.9
Glyphs, banners and distractors	82.6	68.0	0	0

The proposed splitter reaches both reported maxima in four conditions, while the minimal baseline fails in four conditions. The difficult cases remain substituted Roman glyphs and distractors. A separate marker experiment resolves 26 of 30 forms, with bar runs and mixed glyphs causing the unresolved cases. Some bar-run cases can be interpreted as question one, making answer misattribution a concrete risk rather than a hypothetical concern [1].

The detailed normalisation experiment recovers 12 of 14 marker shapes, rejects two, and reports no incorrectly converted number within that small set. This result should not be confused with the broader marker-form test, where misresolution remains. The

distinction illustrates why counts from different experiments must not be merged simply because both concern question numbering.

Table 6.4 presents retrieval outcomes. Seventeen of 18 in-scope queries return context, while all 36 cross-course and 18 off-syllabus queries return none. No foreign-course chunk is reported across the 72 queries. The filter-off ablation matches the in-scope result in the tested index, so the experiment does not demonstrate a marginal performance improvement from the metadata filter [1]. Course filtering still expresses the intended scope constraint, but a harder overlapping-course dataset is needed to measure its discriminating effect.

Table 6.4 Reported retrieval and normalisation outcomes

Test	Denominator	Outcome
Marker resolution	30 forms	26 resolved, 4 unresolved or misresolved
Marker normalisation	14 shapes	12 recovered, 2 rejected, 0 wrong conversions
In-scope retrieval	18 queries	17 returned context; mean distance 0.266
Cross-course retrieval	36 queries	36 returned no context
Off-syllabus retrieval	18 queries	18 returned no context
Foreign-course chunks	72 total queries	0 observed

The training corpus contains 250 examples in each of five answer classes. Mean assigned scores are 7.42 for golden answers, 8.50 for semantic or layman answers, 5.00 for partial answers, 0.50 for confident incorrect answers, and 0.00 for adversarial or invalid input [1]. These are label statistics, not achieved model scores. The higher layman mean than golden mean requires investigation before a controlled claim about adaptation quality.

6.3.2 Diagram detection results

Table 6.5 records the detector configuration and preliminary results [2]. Training uses YOLOv8l with five classes, an image size of 640 pixels, a batch size of eight, and 100 epochs. The paper describes an inference confidence threshold of 0.60. Detection

precision and mAP should remain separate because they summarise different aspects of the detector's output.

Table 6.5 Diagram detector evidence

Item	Reported value
Training images	350
Validation images	5
Object classes	5
Training epochs	100
Image size	640 pixels
Batch size	8
Precision	88.4%
mAP at IoU 0.5	61.3%
Mean detector processing time	37.1 ms per image

The five-image validation set is too small to establish general performance across handwriting styles, layouts, and capture conditions. No confidence interval or independent held-out test result is reported. The paper explicitly identifies OCR accuracy, structural reconstruction, connection detection, and grading agreement as future evaluation. The reported processing time must therefore not be used as the latency of the complete diagram workflow.

At group level, the principal implication is that a detected shape is intermediate evidence. A correctly localised entity may still receive an incorrect OCR label or be associated with the wrong relationship. Conversely, a missed line can invalidate a criterion even when every shape is detected. Full-pipeline validation should record these errors separately so that later rubric disagreement can be traced to its originating stage.

6.3.3 Viva score integrity results

The viva component reports a 20,000-vector simulation with thirteen missingness conditions [3]. Mean absolute score disturbance is 3.04 points under renormalisation and 9.12 under zero-imputation. Mean grade-band changes are 29.5% and 65.8%,

respectively. Table 6.6 includes the correlated-degradation and rank-stability results to preserve the trade-off rather than reporting only the favourable comparison.

Table 6.6 Reported viva missingness comparisons

Measure	Renormalisation	Zero-imputation
Mean absolute disturbance, independent sampling	3.04 points	9.12 points
Mean grade-band change rate	29.5%	65.8%
Mean absolute disturbance, correlated degradation	2.82 points	9.09 points
Grade-band change rate, correlated degradation	26.3%	62.6%
Mean Spearman rank correlation	0.896	0.911

Zero-imputation retains slightly better average rank correlation in the reported independent population, while renormalisation better preserves absolute scores and grade bands. The correct conclusion is a trade-off relevant to fixed thresholds, not universal superiority on every measure. The acoustic-family ablation can increase the renormalised mean because the remaining families score higher in the synthetic population [3]. Renormalisation preserves observed coverage; it does not guarantee that the expected score stays constant.

Residual grade changes are also material. Removing the engagement family changes grade bands for 42.7% of vectors under renormalisation, and removing the acoustic family does so for 60.0% [3]. A lower pooled average is not evidence that individual candidates are unaffected. The system should expose available families and avoid interpreting reduced disturbance as proof of fairness.

The ±10% perturbation experiment reports a maximum single-input movement of 1.52 points and average movement by input no greater than 1.03. Constant sweeps report mean changes no greater than 1.15 and minimum rank correlation 0.9924 [3]. These results concern sensitivity within the selected ranges. The constants remain

engineering choices and the most influential engagement features still require external validation.

6.3.4 Analytics and recommendation results

The classifier matches all eleven temporal-fixture labels, with accuracy and six-class macro-F1 of 1.000 [4]. Five classes have two examples and Evaluate has one. The reported exact normalised comparison finds no matches against the available 8,767-question training CSV. This is a limited overlap check, not proof against paraphrase leakage or undocumented checkpoint exposure.

The recommendation study uses ten synthetic weakness profiles and two project-affiliated raters. Table 6.7 presents the reported means. Precision at five measures the share of selected items marked usable; NDCG at five also accounts for their positions in the ranked list. The two metrics address different aspects of recommendation quality and should be retained together.

Table 6.7 Recommendation comparison across ten profiles

Method	Rater 1 P@5	Rater 1 NDCG@5	Rater 2 P@5	Rater 2 NDCG@5
Random	0.320	0.349	0.291	0.357
Weakness only	0.360	0.436	0.320	0.429
Weakness weight 0.20	0.300	0.320	0.260	0.314
Default weight 0.35	0.360	0.420	0.320	0.414
Weakness weight 0.50	0.360	0.458	0.320	0.452

The default method matches weakness-only precision for both raters, including zero profile-level precision differences. Its mean NDCG differences from weakness-only are -0.016 and -0.015, with reported paired 95% bootstrap intervals of [-0.112, 0.059] and [-0.111, 0.059]. Both intervals include zero [4]. The additional signals therefore do not establish incremental value in this pilot. The improved point estimates at weight 0.50 remain exploratory because the same rated corpus informs the comparison.

Question-bank coverage limits what any ranker can return. The 179 records contain 90 lecture items, 55 tutorial items, 18 generated items, and 16 prior-exam items. Their

cognitive distribution is 132 Apply, 39 Understand, and eight Analyze, with no Remember, Evaluate, or Create items [4]. A six-class classifier does not compensate for absent classes in the bank. Sparse topics, including indexing, transactions, and recovery, further constrain useful practice selection.

The analytics thesis also reports a metric implementation discrepancy. Its general NDCG helper constructs the ideal ranking from the truncated list, while the supplementary reproduction uses the full candidate pool [4]. That difference can change the result and should be resolved before future comparisons. The common report retains the full-pool interpretation and does not claim that the repository helper has been corrected.

6.3.5 Integrated interpretation

Across the four studies, the strongest shared finding is that intermediate evidence and conservative status handling are necessary for accountable assessment support. The written pipeline exposes answer-source and engine labels; the diagram pipeline exposes objects and connections; viva scoring records availability and exclusion; analytics separates numeric evidence from semantic and generated outputs. These mechanisms make errors more inspectable, although their combined effectiveness still requires a group execution study.

The results also show why technical success must not be promoted into an educational claim. Retrieval scope is not grading validity. Detector precision is not graph accuracy. Stability under missingness is not agreement with examiners. Correct labels on eleven questions are not general semantic understanding. A recommendation that a rater would use is not evidence of improved attainment. The common conclusion is therefore one of prototype feasibility and specific engineering findings, with academic and institutional validation still required.

6.4 Stakeholder and user validation

The available user-related evidence is narrow. The analytics study uses two project-affiliated raters who assess 100 candidate rows across ten profiles [4]. They mark 32% and 29% of candidates as usable. Their binary would-use kappa is 0.882, while exact

five-point overall kappa is 0.744. These values describe internal agreement on the study materials, not representative lecturer acceptance of the platform.

The rating dimensions reveal different interpretation problems. Weakness-fit kappa is 0.208, but Spearman correlation is 0.891 and 95% of pairs are within one point. This suggests substantial agreement in ordering alongside different scale anchors. Difficulty-fit kappa is -0.088 and rank correlation is -0.029, despite 98% within-one-point agreement [4]. Difficulty therefore requires a clearer shared definition and anchored examples before it can support a persuasive usefulness claim.

The handwritten thesis states that formal lecturer feedback and agreement with human markers are planned [1]. The viva thesis proposes authentic recorded sessions with independent examiners but reports no completed human-grounded validation [3]. The diagram paper identifies comparison against expert marks as future work [2]. Combining these plans does not produce a completed group stakeholder study.

A suitable next study should recruit lecturers independently of the development team and assess the entire review task. Outcomes should include mark corrections, reasons for rejecting proposals, ability to identify processing failures, time spent reviewing, perceived explanation quality, and suitability of practice recommendations. Student perspectives should address comprehensibility, accessibility, and the review process. Consent and institutional approval should precede authentic-data collection.

The study should compare assisted and ordinary workflows using equivalent material and account for order effects and reviewer learning. Self-reported time savings should be separated from measured task time. High satisfaction should not substitute for correct detection of erroneous results. These distinctions would help determine whether the interface supports critical review or merely makes uncertain outputs appear convincing.

6.5 Achievement of common objectives

Table 6.8 assesses the objectives from Chapter 2 against the current evidence. Partial achievement is appropriate where an implementation or local study exists but the complete required outcome has not been demonstrated. This recognises the work

completed while preserving the difference between a functioning prototype and a validated institutional system.

Table 6.8 Group objective achievement

Objective	Status	Evidence and remaining requirement
O1 Multimodal workflow	Partly achieved	Four complementary component paths documented; controlled group execution outstanding
O2 Shared evidence meaning	Partly achieved	Component provenance and status mechanisms exist; common contract and consumer checks outstanding
O3 Reviewable proposals	Partly achieved	Written review and overrides reported; formal cross-component reviewer validation outstanding
O4 Analytics and practice	Partly achieved	Engine and client connections documented; approval-to-recommendation run outstanding
O5 Appropriate component evaluation	Substantially addressed within stated scope	Baselines and limitations reported; independent and authentic-data validation incomplete
O6 Pilot readiness	Not established	No consolidated human, security, recovery and load evidence

The group aim is addressed as an integrated design and prototype research programme. It is not fully established as a deployable academic assessment service. The most credible achievements are the separation of processing responsibilities, component evidence records, availability-aware scoring, explicit baseline comparisons, and a shared path towards assessment-driven practice. The remaining work concerns correctness across boundaries and validity in the intended educational setting.

Objective achievement should be updated only when corresponding evidence is available. A new diagram accuracy result should not automatically change the status of human grading validity. A successful browser demonstration should not close recovery or fairness requirements. The objective matrix is intended as a living verification record in which each claim is attached to the test or study that actually supports it.

6.6 Integrated limitations and threats

Error propagation is the principal system-level threat. A misread identifier can attach correct grading to the wrong student; a wrong answer boundary can corrupt a rubric result; a missed diagram connection can change interpretation; and a low-quality transcript can affect viva features. Analytics may then amplify the original error by turning it into a weakness profile and repeated practice advice. A review stage reduces risk only if the reviewer can identify the originating evidence and has time to inspect it.

Construct validity is limited across several outputs. Behavioural delivery is not technical mastery, Bloom category is not empirical item difficulty, and observed low achievement is not a validated latent knowledge estimate. Ranking scores are not probabilities of future exam content. The platform should retain these distinctions in both data contracts and interface labels. Otherwise, technically accurate calculations can still support an invalid interpretation.

External validity is limited by synthetic written material, a five-image diagram validation set, simulated viva features, and a single-course analytics pilot with affiliated raters. These studies cannot establish broad performance across institutions, languages, disabilities, handwriting styles, or recording conditions. The absence of authentic student data also means that the observed failure distributions may not resemble those encountered in an examination setting.

Internal validity and reproducibility have additional constraints. The grading corpus has an unexplained class-score inversion; diagram downstream stages lack measured outcomes; viva raw simulation artefacts are not available in the thesis bundle; and the analytics study identifies tie-order and metric-helper differences [1]–[4]. The analytics rating data contain 100 rows but only 91 distinct profile-and-question identifier combinations, which requires an explicit candidate-identity policy in subsequent work [4].

Operational limitations include external quotas, temporary inference hosting, package-name collisions, in-process jobs, and unmeasured group throughput. Privacy and fairness limitations remain unresolved for authentic recordings and scripts. A common interface can make the product appear more mature than its evidence warrants, so

deployment decisions should use the objective matrix and recorded tests rather than visual completeness.

The proposed controls are evidence preservation, explicit statuses, conservative publication, independent datasets, reviewer training, and reproducible releases. These are necessary conditions for stronger validation, not guarantees of success. Further research may show that some features, ranking terms, or automation routes should be simplified or removed if they do not add demonstrable value.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The proposed value of Gradex is to reduce fragmented assessment work while preserving a lecturer's ability to inspect and revise the result. Recognition, answer organisation, retrieval of teaching material, provisional grading, and analysis can be connected through shared context. The benefit is a reviewable first pass and a traceable route to follow-up practice, rather than an unsupported promise that academic judgement can be eliminated.

Lecturers could benefit from reduced repetition and easier access to evidence, while students could receive more specific and timely feedback. Course coordinators could inspect patterns across topics and cognitive levels. Technical teams could reuse component boundaries and versioned contracts. These benefits remain hypotheses where the studies have not measured review time, feedback delay, error reduction, or learning outcomes.

The first credible adoption setting is a supervised departmental pilot that runs alongside ordinary assessment procedures. Existing marks should remain authoritative until the assisted workflow is evaluated and approved. Pilot success should require that reviewers can detect and correct material errors, not merely that the platform completes processing. Analytics and practice suggestions can be evaluated separately from any use that affects official marks.

The evidence suggests that the strongest initial product claims concern workflow and provenance. The written engine demonstrates reported synthetic batch operation, viva scoring formalises a missing-data policy, and analytics exposes useful limitations through baseline comparisons [1], [3], [4]. A commercial proposition should retain those boundaries instead of advertising a single accuracy figure that the studies do not support.

7.2 Market alternatives and differentiation

The principal alternatives are ordinary lecturer marking, digital rubric workflows, standalone recognition services, general language-model tools, specialised diagram

assessors, and separate analytics or recommendation systems. This is a functional comparison, not a current market survey or a claim about the capabilities of every named product. The component reports do not provide a systematic commercial benchmark or customer interview study.

Manual marking offers contextual judgement and flexible interpretation, but requires sustained lecturer effort. Digital workflows can support annotation and moderation without automating every judgement. Standalone OCR can provide text while leaving question alignment and grading unresolved. General-purpose language tools can respond to rubrics, but an institution must still solve identity, scope, provenance, review, and recovery. The group opportunity lies in connecting these responsibilities around assessment evidence.

Table 7.1 describes the intended differentiation at workflow level. Its entries identify responsibilities rather than unsupported claims of competitive superiority. The current prototype's weaknesses remain important: limited authentic-data validation, external dependencies, uneven component maturity, and no measured economic advantage.

Table 7.1 Functional positioning of the integrated solution

Approach	Main responsibility	Remaining integration need
Manual assessment	Examiner interprets evidence and applies rubric	Administrative organisation and timely feedback
Digital rubric workflow	Capture, annotate and moderate marks	Automated extraction and cross-modal evidence if required
Standalone OCR or LLM	Recognition or prompted interpretation	Identity, rubric context, review and failure semantics
Separate analytics tool	Summarise existing records	Reliable source marks and traceable learning action
Gradex prototype	Connect specialised processing, review and practice	Independent validation and dependable operation

The defensible differentiation is the continuity between heterogeneous evidence, review, and derived action. That differentiation must be tested against simpler alternatives. The analytics result is instructive: adding contextual signals did not

establish improvement over weakness alone [4]. A product should therefore retain complexity only where it supports a measured need or an essential evidence boundary.

7.3 Business model and feasibility

An institution-managed service is a plausible initial sustainability model because the system relies on local rubrics, course materials, student records, and academic accountability. A departmental licence, support agreement, or managed deployment could fund maintenance if a pilot establishes value. Usage-based pricing is another possibility, but it should not encourage processing volume at the expense of careful review. No pricing, revenue, adoption, or willingness-to-pay result is reported.

The main cost categories are OCR requests, model inference, video processing, storage, database operation, content curation, support, and periodic evaluation. Human review is part of the operating cost rather than a temporary inconvenience to ignore. A system that produces many plausible but incorrect proposals may increase total work even if inference is inexpensive. Feasibility should therefore compare total task effort and correction burden with ordinary practice.

Technical feasibility is partly supported by the reported prototype workflows and component experiments. Operational feasibility remains constrained by notebook hosting, changing endpoints, in-process jobs, and absent group load measurements [1]–[4]. Moving to stable model hosting and recoverable job execution should follow a measured pilot workload rather than an assumed scale target.

Content maintenance is equally important. Analytics depends on reviewed question-bank coverage, and the current bank omits three Bloom categories [4]. The grading adapter depends on label quality, and the corpus inversion requires audit [1]. A sustainable service needs ownership for updating rubrics, approving generated questions, reviewing labels, and reevaluating changed models. Hosting alone cannot sustain educational quality.

Commercial progression should therefore use staged evidence gates: a technically reproducible release, a supervised workflow study, independent assessment validation, a documented operating cost, and an institutional adoption decision. These stages are a

proposed feasibility path. The project should not represent them as completed commercial milestones.

7.4 Adoption intellectual property and compliance

Adoption depends on trust grounded in inspectable evidence. Lecturers need to know when to rely on an automated proposal, when to correct it, and when to decline processing. Students need understandable information about the use of their work and a meaningful review route. Training should include examples of confidently wrong outputs, missing evidence, and service failures so that users do not equate polished explanations with correctness.

The ownership and permitted reuse of source code, trained adapters, generated training material, lecture resources, and question-bank content require confirmation with the institution and contributors. The model, dataset, and service licences should be reviewed against the intended deployment. The common report does not assert that any particular commercial use is licensed merely because a model or library is accessible.

Data governance must address the complete flow. Script images and diagram crops can be sent to external OCR; grading may use a hosted fallback; viva recordings contain identifiable voice and facial information. Access, retention, deletion, processor arrangements, and permitted research reuse should be resolved before authentic material is processed. Applicable legal and institutional requirements need a deployment-specific review; no formal compliance claim is made here.

Human oversight should be operational rather than symbolic. Approval should require access to the evidence and preserve the reviewer's changes. An appeal should be able to recover the relevant rubric, input, model version, and correction history. If a component's feature set cannot be justified for the intended construct or population, removing that signal may be more appropriate than adding an explanation to its score.

The UNESCO guidance cited in the component literature places educational use within a human-centred research and governance context [12]. For Gradex, that context means evaluating both system behaviour and how people use the output. A responsible release includes uncertainty, failure handling, and accountable review as core product behaviour rather than as a disclaimer attached after processing.

7.5 Sustainable Development Goal contribution

A plausible contribution to SDG 4 requires evidence that the platform improves the assessment and feedback process without introducing unacceptable disadvantages [7]. Suitable indicators include time to approved feedback, the proportion of feedback judged useful, correction rates, and access to appropriate practice. None has yet been established as a group-level outcome. They should form part of the next supervised pilot rather than being inferred from component accuracy.

Equity should be evaluated through meaningful conditions and groups. Handwriting clarity, language background, assistive needs, recording quality, and device access may change error rates or completion. Under the reported simulations, the viva policy reduced the artificial downward score shifts associated with zero-imputation. The validity and fairness of the permitted features still require independent evaluation [3]. The written and diagram routes require similar scrutiny of capture conditions and reviewer escalation.

Resource sustainability requires measurement of model calls, processing time, storage growth, and retained media. Reusing validated cached results can reduce repeated computation, but cache invalidation must follow changes to models, taxonomies, or rubrics. Retention limits can reduce both storage cost and exposure of personal evidence. These controls should be assessed alongside usability because overly restrictive deletion can also make legitimate review impossible.

The group's contribution to innovation is strongest where its methods are documented and their limits are visible. Reproducible failure analysis can be useful even when a complex technique fails to outperform a simpler baseline. The intended long-term benefit is an assessment-support process that remains understandable, maintainable, and open to correction as educational needs and evidence change.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

Gradex addresses a shared assessment problem through four complementary components: handwritten answer grading, entity relationship diagram evaluation, viva score integrity, and assessment-driven practice recommendation. The common architecture connects evidence acquisition, specialised interpretation, human review, and derived learning action. Its central principle is that a result should retain the evidence and status required to understand how it was produced.

The current research supports selected engineering findings. The handwritten pipeline handles several synthetic marker conditions and reports course-scoped retrieval without observed foreign-course leakage in its test set. The diagram detector demonstrates preliminary feasibility on a very small validation set. Reported viva simulations show that renormalisation reduced mean absolute score disturbance from 9.12 to 3.04 points and grade-band changes from 65.8% to 29.5%, with the same direction of improvement under correlated degradation [3]. Analytics demonstrates a limited six-class fixture result while showing no established ranking advantage from the default rule's additional signals [1]–[4].

These findings do not establish the validity of autonomous grading or the effectiveness of the complete platform in an authentic institutional setting. The studies use different domains, datasets, and outcome measures, and no single controlled group execution combines all relevant processing and review boundaries. Human-marker agreement, independent stakeholder validation, and operational readiness therefore remain unresolved.

The overall problem is addressed provisionally through a connected design and prototype evidence. The strongest conclusion is that meaningful integration requires more than a shared interface: identity, rubric scope, missingness, provenance, approval, and correction must survive every boundary. A result that loses those properties can mislead downstream analytics even when the consumer's calculations are numerically correct.

8.2 Principal integrated contribution

The principal integrated contribution is an assessment-to-practice architecture in which distinct forms of student evidence retain their own processing logic and interpretation. Written answers provide transcript and rubric evidence; diagrams provide structure; viva analysis provides bounded behavioural features and availability; analytics connects observed achievement to reviewed practice. The design supports a common workflow without assuming that all outputs measure the same construct.

A second contribution is the common evidence model developed through synthesis. Processing success, academic score, diagnostic information, review status, and generated advice require distinct treatment. The difference between a missing value and an observed low value is visible in viva scoring, but applies more broadly to OCR failures, emergency grading responses, and absent question-bank coverage. This shared interpretation provides concrete requirements for the next integration release.

A third contribution is an evidence-bounded evaluation account. The group report preserves the small samples, synthetic populations, baseline trade-offs, and incomplete validations rather than producing a misleading combined accuracy. Negative or inconclusive findings are useful: the recommendation comparison questions unnecessary ranking complexity, while the source inconsistencies identify artefacts that must be reconciled for reproducibility.

The product contribution and research contribution remain related but distinct. A usable workflow can exist before its academic decisions are validated, and a sound local experiment can exist before its component is operationally integrated. The objective matrix identifies both kinds of progress and the evidence still required to connect them.

8.3 Future directions

The first priority is to complete the shared contract and a controlled integration run. The team should reconcile identity fields, rubric versions, score scales, statuses, and approval semantics, then execute the cases in Appendix C. Particular attention should be given to emergency zeros, incomplete viva results, repeated submissions, reviewer

corrections, and stale analytics. The result should be a versioned evidence package containing requests, responses, screenshots, expected outcomes, and test logs.

The second priority is to address known component defects and uncertainty. The handwritten marker parser should be checked against bar runs and mixed glyphs, and the synthetic training labels should be audited. Diagram evaluation needs larger independent detection data and measurements of OCR, connection, graph, and expert-grade accuracy. Viva work should package the seeded generator, scoring-policy version, raw aggregate outputs and test logs, then proceed to authentic-recording calibration and independent examiner ratings. Analytics needs a corrected metric contract, explicit tie policy, stronger topic labels, and broader bank coverage.

The third priority is independent academic validation. Authentic scripts, diagrams, and recordings should be collected only under an approved protocol, with qualified raters using declared rubrics. Inter-rater reliability should be established before interpreting agreement with the system. Evaluation should include different capture conditions and relevant participant groups, and separate behavioural delivery from technical knowledge. The primary and fallback graders should be compared on the same cases.

The fourth priority is evaluation of review and learning action. Independent lecturers should assess whether evidence displays help them identify errors and whether the workflow reduces total effort after corrections. Practice ranking should be tested on held-out profiles with calibrated raters and a strong simple baseline. Only after content and ranking quality are adequate should a student study investigate learning outcomes.

Finally, the team should establish an operational release process with stable hosting, recoverable jobs, protected configuration, access controls, backup and restore, and measured workload behaviour. Commercial and institutional adoption should follow those results. The immediate objective is a reproducible supervised pilot whose benefits and limitations are visible to everyone responsible for assessment decisions.

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APPENDICES

Appendix A Integrated interface specification

This appendix defines a proposed logical contract for validating the group workflow described in Chapter 4. It does not claim to reproduce an implemented universal JSON schema. Component owners should map their actual fields to the contract, resolve disagreements, and version the agreed representation before the controlled integration run.

A.1 Identity and scope

Every processing request should identify the course, assessment, rubric version, student or unresolved-identity state, submission, and component route. The originating file should have a stable evidence reference. A change in the inferred student identity should require a recorded correction rather than silently overwriting the previous association. The consuming service should reject inconsistent course and rubric references.

Table A.1 Proposed shared record semantics

Logical field	Meaning	Validation rule
Course and assessment identity	Educational scope	Must resolve to authorised records
Student identity	Person associated with work	Verified or explicitly unresolved
Submission identity	One submitted evidence item	Stable across retries
Rubric version	Criteria and maxima used	Must remain traceable after edits
Component and version	Processing route	Identifies code and model policy
Evidence reference	Input and intermediate artefacts	Accessible to authorised reviewers
Processing state	Success, partial, failed or pending	Independent of numeric score
Review state	Unreviewed, approved or corrected	Independent of processing success
Score and scale	Proposed or approved numerical value	Range and denominator explicit
Superseded result	Link to earlier output	Retains correction history

A.2 Score and status policy

An unavailable score should be represented as absence with a reason, not coerced to zero for convenience. Existing component emergency responses may carry a number, but a consumer should consider the associated state before treating it as academic evidence. A genuine rubric zero, a missing answer, and a failed model call are different events even when their display values coincide.

Mode A viva results should retain the behavioural construct and available-family set. Mode B should retain the external technical score and its scale, together with the fusion policy. If T is absent, the fused result remains pending. Diagram and written results should retain criterion maxima and reconstruction or transcript evidence. Recommendation scores should be typed as ranking quantities rather than assessment marks.

A.3 Review and correction

An approval record should identify the reviewer, decision time, result version, and any changes. Corrected marks should not erase the automated proposal. Corrections to evidence or rubric interpretation should invalidate affected analytics, and the next analysis should identify the approved input version. Unresolved source errors should remain visible in coverage summaries rather than disappearing from denominators without explanation.

A.4 Boundary examples

Example 1 concerns a failed grader. Both inference routes time out, so the written engine returns its flagged emergency response. The shared consumer records processing failure and excludes the value from approved academic totals. The lecturer can retry or mark manually. This example specifies desired consumer behaviour; its implementation requires the integration test in Appendix C.

Example 2: When the viva acoustic family is unavailable but integrity gates pass, renormalisation may return a behavioural score from the remaining families with reduced coverage. This is not equivalent evidence to a complete recording.

Appendix B Team contribution declaration

The contribution matrix in Section 4.5 records the responsibilities described in the component reports. Each member should confirm their own contribution and the agreed shared responsibilities before signing. The attribution of overall client design and grading-workflow implementation should be reconciled explicitly so that neither contribution is erased or counted twice.

Table B.1 Team confirmation record

Student ID	Component	Signature	Date
IT22239198	Handwritten answer grading		
IT22262554	ER diagram evaluation		
IT22557056	Viva score integrity		
IT22134776	Analytics and practice recommendation		

Shared contributions to confirm include architecture decisions, interface agreement, common client integration, group testing, final evidence review, and preparation of this report. Signing confirms the agreed attribution; it does not independently reproduce the numerical experiments or replace the supervisor's certification.

Generative AI assisted the synthesis, language editing, organisation, and formatting of this common report from the four supplied component documents and the supplied reporting template. The team must review the resulting interpretations, numerical transcriptions, references, and contribution statements. No new experimental measurements were generated for the common report, and AI assistance does not establish the validity of the reported research findings.

Appendix C Integrated acceptance test protocol

This prospective protocol supports the evaluation framework in Chapter 6. All cases require an identified release, fixed inputs, expected output, execution log, and reviewer. Their status is not executed within this common-report synthesis. Passing a case establishes the stated behaviour under its test conditions; it does not establish academic validity beyond those conditions.

Table C.1 Proposed complete-system acceptance cases

ID	Scenario	Required outcome
INT01	Valid written script and roster identity	Same student, course and rubric throughout review and storage
INT02	Duplicate or unreadable student identity	No silent attribution to a guessed student
INT03	Diagram-only page	Retained as diagram evidence rather than assumed blank academic work
INT04	Mixed text and diagram assessment	Separate processing with traceable question association
INT05	Primary grading endpoint fails	Fallback provenance retained per question
INT06	Both grading routes fail	Failure visible and excluded from approved analytics
INT07	Course mismatch in retrieved context	Out-of-course material rejected
INT08	Missing viva feature or family	Correct omission, denominator and coverage record
INT09	Failed viva integrity gate	No unsupported numerical result
INT10	Mode B without technical score	Pending fusion retained through client and storage
INT11	Reviewer changes a mark	Audit record and recalculated analytics use corrected value
INT12	Reviewer changes a topic	Old practice output invalidated or recomputed
INT13	Repeat request after interrupted response	No duplicated approved assessment evidence
INT14	Server restart during a batch	Documented recovery or explicit incomplete-job state
INT15	Unauthorised course or evidence access	Access denied without exposing protected content
INT16	Missing classifier or generator	Component-specific failure state accurately displayed

The test dataset should include ordinary cases and deliberately difficult cases. Text cases should cover varied markers, bar runs, missing questions, and embedded instructions. Diagram cases should cover crossing lines, nearby shapes, weak labels, and missing connections. Viva fixtures should distinguish true silence from transcription failure and should mutate excluded diagnostics while holding permitted features fixed. Analytics fixtures should include sparse coverage, unclassified topics, ties, repeated identifiers, and corrected source marks.

Each case should compare stored results with displayed results, because a backend flag is insufficient if the interface omits it. The evidence package should include the actual request and response, selected database fields, and a screenshot or equivalent review record. Sensitive authentic data should be excluded from public artefacts or appropriately de-identified under the study protocol.

Performance testing should use a declared workload rather than only isolated successful requests. Measure completion time, queue delay, service retries, memory use, and incomplete outcomes at increasing concurrency. Report the hardware, hosted-service limits, and whether models are warm or cold. Academic correctness and recovery semantics must remain intact under load; faster output alone is not a passing result.

Human validation should be a separate study with qualified independent raters and a written rubric. Establish inter-rater reliability, then compare the system and reviewed workflow with those judgements. Report disagreements by stage and criterion. For recommendations, use held-out profiles and predeclared candidate identity, relevance, and tie-handling rules. Do not tune weights on the same cases used to claim final improvement.

Appendix D Research evidence and reconciliation record

This appendix identifies the evidence required to strengthen the group claims. It supports Sections 5.4 and 6.6 by separating an unresolved research artefact from a missing presentation detail. The table is a completion record, not a claim that the listed materials have already been collected.

Table D.1 Evidence to archive or reconcile

Component or area	Required material	Purpose
Handwritten grading	Exact segmentation metric definitions and original outputs	Interpret Score 1 and Score 2 without guesswork
Handwritten grading	Reconciled 14-case description and marker results	Resolve abstract versus detailed-results inconsistency
Grading model	Dataset checksum, adapter checksum and label audit	Establish training provenance and class-score meaning
Diagram evaluation	Independent test split and stage-level results	Separate detection, OCR, graph and grading performance
Viva evaluation	Seeded generator, scoring version and test logs	Reproduce simulation and invariance claims
Analytics	Exact eleven-question fixture and model provenance	Support classification and overlap statements
Recommendation	Full-pool NDCG implementation and tie policy	Reproduce reported ranking metrics consistently
Recommendation data	Candidate-identity and duplicate-row decision	Clarify 100 rows versus 91 identifier combinations
Group integration	Versioned schemas and INT01–INT16 logs	Demonstrate shared workflow and recovery
Team execution	Agreed contributions and available project records	Support attribution and shared execution
Authentic evaluation	Approvals, protocols and independent ratings	Establish academic and stakeholder validity

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